

Mission-Driven Corner-Speed Design of a Low-Voltage Switched Reluctance Motor Drive for Electric Bicycles

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Abstract—This paper presents an analytical, finite-element (FE) and simulation-based design methodology for a switched reluctance motor (SRM) drive for a 250-W, 25-km/h pedelec supplied from a 36-V battery. The vehicle mission is translated into motor torque–speed requirements, and the phase turns are chosen at a mission-derived *corner speed* rather than at the assist cut-off speed. The resulting relation $P_c = k_c V_{dc} I_{pk}$, a sizing form of the classical converter volt-ampere argument, is tested quantitatively. Across nine designs spanning a 3.9-fold range of active mass, the required peak current shows no mass dependence ($R^2 = 0.004$) and $k_c = 0.60 \pm 0.03$. Across 24-, 36- and 48-V redesigns, the current scales as $V^{-1.00}$. A 12/8 machine is sized analytically and its characteristics are computed with a nonlinear 2-D FE model. This numerical verification shows errors of the analytical model of up to 23% in flux linkage and 17% in mean torque, so FE tables are used for all drive-level predictions. Three-phase FE solves confirm that mutual coupling changes mean torque by at most 1.5%. The predicted performance of the $\varnothing 135$ -mm, 3.4-kg (active) machine with an 8:1 gear is as follows. It meets the 5.93-N·m launch torque at 26.5 A with an 11.6% margin at any battery voltage between 30 and 42 V. It reaches 80.2% motor and 77.9% battery-to-shaft efficiency at the rated point and consumes 2.5–10.9 Wh/km on flat, hilly and aggressive missions. Cubic torque-sharing control cuts torque ripple from about 197% to 10–33% below 500 r/min, at a cost of 7–12 efficiency points. An equal-envelope analytic surface-PM benchmark is about 12 points more efficient at the rated point and uses less energy on hilly and aggressive missions. The SRM uses less energy on the flat light-assist mission, needs no magnets and has no drag when coasting. No prototype measurements are reported, and all performance figures are model predictions.

Index Terms—Asymmetric half-bridge converter, electric bicycle, finite-element analysis, light electric vehicle, machine design, pedelec, switched reluctance motor (SRM), torque-sharing function, torque ripple.

I. INTRODUCTION

ELECTRIC bicycles are the most widely sold electric vehicles worldwide. Their drives are almost exclusively brushless permanent-magnet (PM) hub or mid-drive motors. PM machines offer high torque density and efficiency, but they rely on rare-earth magnets. Those magnets carry supply-chain and price risk, limit operating temperature and add drag torque when unpowered (cogging and eddy-current losses).

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Source code, FE scripts and all data used in this paper are available at <https://github.com/subrahmanyamtanala-rgb/Analysis-and-Design-Considerations-for-a-srm-based-electric-drive> (tag v2.0-rev2).

This last point matters for a pedelec, which by regulation must be pedaled and is ridden unassisted above 25 km/h [1].

The switched reluctance motor (SRM) addresses these concerns. Its rotor is a stack of laminations, its stator carries concentrated coils, and its phases are electrically and magnetically independent to first order. This gives inherent fault tolerance and zero torque when the windings are not excited [2], [3]. SRMs have been studied extensively for electric and hybrid vehicles [4]–[9]. Their known drawbacks are torque ripple, acoustic noise and lower efficiency at small power ratings [10]–[12]. Classical design procedures [13]–[15], pole-number studies [16] and geometry-optimization methods [17] exist, but most of them target kilowatt-class machines with DC-link voltages of several hundred volts. Light-vehicle work has addressed torque control [18]. For e-bikes specifically, Terzić and Mihić designed, built and tested a 6/10 SRM for a mid-drive replacing a commercial PM machine [19]. The 250-W, 36-V hub-drive regime combines a severe volt-second limit, a launch torque about four times the rated torque, a pancake hub envelope and a mission dominated by part-load cruising. To the author’s knowledge, the combination of mission-derived corner-speed turn selection, FE-based current sizing, and converter, thermal and control evaluation has not been reported systematically for a 250-W, 36-V geared SRM pedelec drive.

This paper contributes the following:

- 1) A mission-driven corner-speed rule for choosing the turns of a low-voltage SRM. Its consequence $P_c = k_c V_{dc} I_{pk}$ is positioned relative to established volt-second and converter volt-ampere arguments and is tested quantitatively against mass (nine designs) and design voltage (24/36/48 V).
- 2) A mission-to-converter design flow that combines analytical sizing with nonlinear 2-D FE numerical verification (including three-phase coupled solves), FE-based current sizing and commutation-angle optimization.
- 3) FE-based performance, loss, converter, DC-link and transient thermal predictions, with sensitivity to battery voltage and state of charge, iron-loss model, PWM frequency, gear efficiency, heat-transfer coefficient, ambient temperature, air gap and pole arcs, evaluated on three standardized missions.
- 4) A quantified comparison of current-reference and torque-sharing-function (TSF) control, and an equal-envelope, equal-boundary comparison with an analytic surface-PM benchmark.

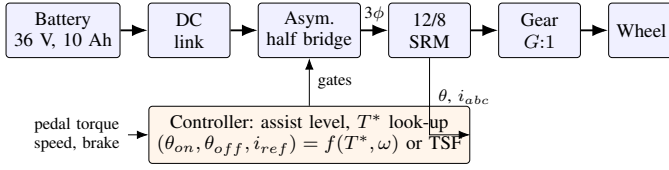


Fig. 1. Architecture of the SRM-based pedelec drive.

The scope is deliberately limited to analysis, FE simulation and system simulation. No hardware results are reported, and all efficiency, range and temperature figures are predictions under stated assumptions (Section X-D). All code, FE scripts and data are openly available [20].

II. SYSTEM DESCRIPTION AND DRIVE REQUIREMENTS

A. Drive Architecture

Fig. 1 shows the drive system. A 10-cell (36 V nominal, 30–42 V) Li-ion battery feeds a three-phase asymmetric half-bridge converter through a DC-link capacitor bank. The SRM drives the rear wheel through a single-stage planetary gear of ratio G inside the hub. The controller uses rotor position, phase currents, pedal-torque or cadence sensing and a speed limit to command assistance up to the 25-km/h cut-off.

B. Road-Load and Gear Models

The tractive force at the tire contact patch is

$$F = C_{rr}mg \cos \alpha + mg \sin \alpha + \frac{1}{2}\rho C_d A v^2 + k_m m \dot{v}, \quad (1)$$

where $m = 100$ kg (80-kg rider plus 20-kg bicycle), $C_{rr} = 0.006$, $C_d A = 0.5$ m² (upright rider), $k_m = 1.05$ accounts for rotating inertia, and $\alpha = \arctan(\text{grade})$. With wheel radius $r_w = 0.33$ m, the motor speed is $\omega_m = Gv/r_w$. The gear is modeled with a load-dependent mesh efficiency η_g and an optional load-independent drag loss P_{nl} proportional to speed, i.e., a constant drag torque $T_{nl} = P_{nl}/\omega_b$:

$$T_m = \frac{Fr_w}{G\eta_g} + T_{nl}. \quad (2)$$

The baseline uses $\eta_g = 0.95$ and $P_{nl} = 0$. Section VIII reports the sensitivity to both parameters.

C. Duty Points

Table I lists the motor-side requirements for $G = 8$. Cruising on level ground at 25 km/h needs only 149 W. Launching on an 8% grade at 0.5 m/s² without rider effort, the worst case for throttle or walk-assist use, needs 5.93 N·m. This is four times the rated torque.

D. Corner-Speed Design Rule

In single-pulse operation, the peak flux linkage a phase reach is limited by the volt-seconds applied during the dwell angle θ_d :

$$\psi_{pk} = \frac{V_{dc} \theta_d}{\omega}. \quad (3)$$

TABLE I
MOTOR-SIDE DRIVE REQUIREMENTS ($G = 8$, $\eta_g = 0.95$)

Duty point	Condition	Torque (N·m)
Cruise	level, 25 km/h (149 W)	0.88
Rated	250 W at 25 km/h	1.48
Hill	6% grade, 15 km/h	3.03
Launch	8% grade, 0.5 m/s ²	5.93
Base speed n_b	25 km/h	1608 r/min
Corner speed n_c	P_{pk}/T_{pk} , $P_{pk} = 500$ W	805 r/min
Max. speed	$1.2 n_b$	1930 r/min

The number of turns per phase follows from $\psi_{pk} = T_{ph} B_{sp} A_p$, where A_p is the stator-pole face area and B_{sp} the design pole flux density. The average torque is $T = (mN_r/2\pi)W_c$, where W_c is the energy converted per stroke, i.e., the area of the ψ - i loop. With a loop-utilization factor $k_c = W_c/(\psi_{pk} I_{pk})$ and θ_d equal to the stroke angle $\varepsilon = 2\pi/(mN_r)$, the power at the speed ω_c where the turns are chosen is

$$P_c = T_{pk} \omega_c = \frac{mN_r}{2\pi} k_c \psi_{pk} I_{pk} \omega_c = k_c V_{dc} I_{pk}. \quad (4)$$

The turns are chosen at the corner speed $n_c = P_{pk}/T_{pk} = 805$ r/min (12.5 km/h) with $P_{pk} = 500$ W, because the launch torque is only needed at walking speed. Above n_c the drive operates at approximately constant power. Under the assumptions of (4), which are a one-stroke dwell, a fixed V_{dc} , the chosen corner power and a similar k_c , choosing the turns for full launch torque at the 25-km/h base speed would instead require $P_c = T_{pk} \omega_b \approx 1$ kW. This is twice the corner power, and therefore approximately twice the peak current. Section IX tests the rule quantitatively and finds $k_c = 0.60 \pm 0.03$ across nine design variants.

E. Relationship to Existing Volt-Second and Energy-Based Methods

Equation (4) is built from established ingredients. The volt-second limit (3) is the basis of every SRM single-pulse analysis [2], [3]. The energy-conversion-loop picture underlies Miller's energy ratio and the classical argument that the converter volt-ampere rating per unit output power is fixed by how well the ψ - i loop is utilized [2]. In that sense, (4) is the converter volt-ampere argument rewritten in sizing form, and we do not claim it as a new law. The established design procedures [13]–[16] differ in how they are used. They size the machine from the output equation (5), which contains the magnetic and electric loadings B and A but not the supply voltage, and then choose the turns at the rated or base speed.

What this paper adds is how the relation is used, which matters at low voltage:

- The turns are chosen at a corner speed derived from the mission (P_{pk}/T_{pk}), not at the speed limit.
- Rearranged as $I_{pk} = P_c/(k_c V_{dc})$, the relation shows that the converter current is fixed before the machine is sized, and that enlarging the machine reduces copper loss but not current. Section IX confirms this quantitatively: the

TABLE II
COMPARISON OF POLE CONFIGURATIONS AT 1930 R/MIN

N_s/N_r	m	Stroke	Strokes/rev	f_{ph} (Hz)	Switches
6/4	3	30°	12	129	6
8/6	4	15°	24	193	8
12/8	3	15°	24	257	6
16/12	4	7.5°	48	386	8
24/16	3	7.5°	48	514	6

required current shows no mass dependence over a 3.9-fold range of active mass and scales as $V_{dc}^{-1.00}$ when the design voltage is changed.

- k_c is reported for this machine class, which allows a first converter rating directly from vehicle data.

III. TOPOLOGY SELECTION AND MACHINE DESIGN

A. Pole Configuration

Table II compares candidate configurations at the maximum speed. The 12/8 machine is selected under the stated envelope, frequency, converter and performance constraints; it is not the result of a global topology optimization. It has the same number of strokes per revolution as the four-phase 8/6 but needs only six switches. Its two pole pairs per phase balance the radial forces and shorten the flux paths, which suits the short, large-diameter hub envelope. Its phase frequency (257 Hz) keeps iron losses moderate with 0.35-mm laminations, whereas 16/12 and 24/16 machines [7], [16] double this frequency at no benefit for a 250-W machine. The 6/4 machine has a 30° stroke, which causes large torque dips and makes starting on a grade more difficult.

B. Main Dimensions

In torque form, the SRM output equation [3], [13] reads

$$T = \frac{\pi}{4} k_e k_d k_2 B A D^2 L, \quad (5)$$

where D is the bore diameter and L the stack length. The coefficients are $k_e = 0.85$, $k_d = 0.9$, $k_2 = 0.7$, $B = 1.7$ T and $A = 45$ kA/m (short-term peak electric loading). For $T_{pk} = 5.93$ N·m this gives $D^2 L = 184$ cm³. An aspect ratio $L/D = 0.5$ and a split ratio $D/D_o = 0.53$ give $D = 71.7$ mm, $L = 35.9$ mm and $D_o = 135.3$ mm. We use (5) only as a starting point; the current required to meet T_{pk} is determined later with the FE characteristic (Section V).

C. Pole Arcs

The stator and rotor pole arcs must satisfy the feasibility conditions of Lawrenson *et al.* [21]:

$$\min(\beta_s, \beta_r) \geq \varepsilon, \quad \beta_s \leq \beta_r, \quad \beta_s + \beta_r < \frac{2\pi}{N_r}. \quad (6)$$

We select $\beta_s = 15^\circ$ and $\beta_r = 17^\circ$, close to the lower-left vertex of the feasible region in Fig. 2(b). This choice maximizes slot area and the inductance ratio. Table III shows the FE-computed sensitivity at constant turns. Widening both arcs by 2° (17/19) raises the mean motoring torque at 30 A by 9.7% but raises L_u by 14% and takes slot area from

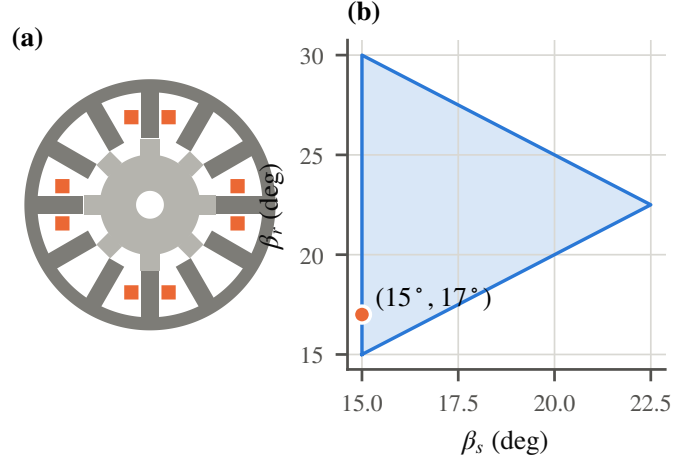


Fig. 2. (a) Lamination of the designed 12/8 SRM with phase-A coil sides marked. (b) Feasible pole-arc region (6) and the selected point.

TABLE III
FE POLE-ARC SENSITIVITY (SAME TURNS): MEAN MOTORING AND PEAK STATIC TORQUE

β_s/β_r (deg)	$\bar{T}_{15A}$ N·m	$\bar{T}_{30A}$ N·m	$\hat{T}_{30A}$ N·m	L_a mH	L_u mH	L_a/L_u
15/17	1.80	4.52	8.39	8.56	1.52	5.64
15/15	1.78	4.51	8.54	8.41	1.45	5.78
15/19	1.79	4.47	8.51	8.62	1.60	5.40
16/18	1.87	4.76	8.55	8.85	1.61	5.48
17/19	1.94	4.96	8.66	9.11	1.73	5.28
15/22	1.75	4.28	8.19	8.66	1.75	4.94

the winding. Widening only β_r to 22° lowers torque. A full co-optimization of pole arcs and winding is left to future work [17]. The baseline is kept because it maximizes copper area for this low-voltage, copper-loss-dominated design.

D. Poles, Yokes, Turns and Winding

The stator yoke is $y_s = 0.75 w_s$, where $w_s = 9.4$ mm is the stator-pole width. The pole flux divides in two in the yoke, so $0.5 w_s$ is the minimum. The extra margin limits yoke saturation and stiffens the stator against the ovalizing vibration mode [2]. The rotor-pole height is limited to $30g$ ($g = 0.3$ mm). With (3) at n_c , the minimum battery voltage of 30 V and a one-stroke dwell, $\psi_{pk} = 93.2$ mWb. This gives $T_{ph} = 164$ turns, i.e., 41 turns on each of the four poles of a phase. Bobbin-wound concentrated coils with a net fill factor of 0.45 use 1.66-mm wire, and $R_{ph} = 0.198 \Omega$ at 100 °C. Table IV summarizes the design. All masses are *active electromagnetic* masses (laminations and copper), excluding housing, shaft, bearings and gear.

E. Analytical Inductance Estimates

The aligned unsaturated inductance follows from two air gaps in series per pole pair, with all $p = N_s/m$ poles of a phase in series:

$$L_a = k_{fe} \frac{\mu_0 (w_s + 2g) L T_{ph}^2}{p g}, \quad (7)$$

TABLE IV
MAIN PARAMETERS OF THE DESIGNED 12/8 SRM

Outer diameter	135.3 mm	Turns/pole	41
Bore diameter	71.7 mm	Wire diameter	1.66 mm
Stack length	35.9 mm	R_{ph} (100 °C)	0.198 Ω
Air gap	0.30 mm	L_a (FE / analyt.)	8.56 / 9.06 mH
β_s / β_r	15° / 17°	L_u (FE / analyt.)	1.52 / 1.37 mH
Stator pole width	9.36 mm	End-winding L_{ew}	0.15 mH
Stator pole height	24.8 mm	Iron mass (active)	2.31 kg
Rotor pole height	9.0 mm	Copper mass	1.09 kg
Stator/rotor yoke	7.0/19.1 mm	Rotor inertia	4.1×10^{-4} kg·m ²

where $k_{fe} = 0.9$. The unaligned inductance combines the path into the rotor slot ($g + h_r$) with two side paths (l_s) and an end-winding factor $k_{ew} = 1.35$:

$$\frac{L_u}{L_a} = \frac{k_{ew}}{k_{fe}} \left(\frac{g}{g + h_r} + \frac{g}{l_s} \right). \quad (8)$$

These estimates are compared with nonlinear FE in Section IV-B.

IV. ELECTROMAGNETIC MODELING AND FE-BASED NUMERICAL VERIFICATION

A. Analytical Saturable Flux-Linkage Model

For fast design iteration, the aligned magnetization curve is represented by an exponential knee model in the spirit of [22], [23]:

$$\psi_a(i) = L_{sat} i + \psi_k \left(1 - e^{-(L_a - L_{sat})i/\psi_k} \right), \quad (9)$$

with $\psi_k = B_k A_p T_{ph}$ and $B_k = 1.55$ T. It is blended with the linear unaligned characteristic by a C^1 position function $w(\theta) \in [0, 1]$:

$$\psi(i, \theta) = L_u i + w(\theta) [\psi_a(i) - L_u i], \quad (10)$$

which gives closed-form co-energy and torque, $T = \frac{dw}{d\theta} [W'_a(i) - \frac{1}{2} L_u i^2]$.

B. Nonlinear 2-D FE Model

The full 12/8 cross-section is modeled with the magnetic vector potential A_z on first-order triangles:

$$\nabla \cdot (\nu(|\mathbf{B}|^2) \nabla A_z) = -J_z, \quad A_z = 0 \text{ on } r = D_o/2, \quad (11)$$

solved by Newton iteration (relative residual $< 10^{-6}$, typically 3–12 iterations) with a monotone PCHIP fit of the M270-35A DC magnetization curve. The geometry is meshed for every rotor position, with an element size of $g/3$ in the air gap growing to 1.2–1.5 mm in iron and air (about 2.9×10^4 nodes). Flux linkage is computed from the coil-side averages of A_z , and an analytical end-winding term $L_{ew} = p \mu_0 N_c^2 l_{ew} \lambda_{ew}$ with $\lambda_{ew} = 0.5$ is added. Torque is computed by Arkkio's method over the whole air-gap annulus [24]. The table covers 16 rotor positions (0–22.5°, mirrored to a full pitch) and 15 currents (1–42 A).

Table V shows mesh convergence. Halving the gap element size from $g/3$ to $g/5$ changes flux linkage by less than 0.1% and torque by 0.03%. As an independent consistency check, the torque obtained by differentiating the FE co-energy agrees with the Arkkio torque to within 0.5% in mean value.

TABLE V
FE MESH CONVERGENCE (15 A)

Gap element (mm)	Nodes	ψ aligned (mWb)	ψ 11° (mWb)	ψ unal. (mWb)	T 11° (N·m)
0.15	16 039	94.45	53.28	20.52	3.169
0.10	29 428	94.53	53.39	20.57	3.168
0.06	53 916	94.57	53.43	20.60	3.169

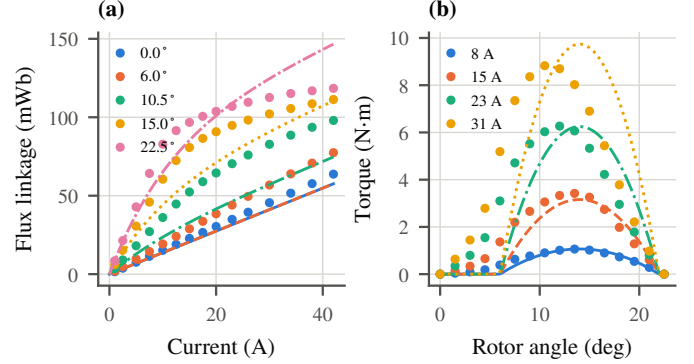


Fig. 3. Analytical model (lines) versus 2-D FE (markers). (a) Flux linkage at five rotor positions (0° unaligned, 22.5° aligned). (b) Static torque of one phase at 8, 15, 23 and 31 A.

TABLE VI
ERROR OF THE ANALYTICAL MODEL RELATIVE TO 2-D FE

i (A)	ψ RMS (%)	ψ max (%)	ψ align. (%)	$\bar{T}$ (FE) (N·m)	$\bar{T}$ err. (%)	$\hat{T}$ err. (%)
8	13.6	19.0	-19.0	0.57	-12.0	-0.1
15	15.3	22.5	-11.8	1.80	-17.4	-8.0
23	13.3	22.5	+1.8	3.30	-10.7	-0.7
31	13.3	22.6	+11.7	4.71	-2.4	+9.9

ψ errors in % of the FE maximum at that current; $\bar{T}$: mean over 0–22.5°; $\hat{T}$: peak. L_a : +5.7%, L_u : -9.7% (analytical vs. FE at 2.5 A).

C. Analytical Model Versus FE

Fig. 3 and Table VI compare the analytical model with FE at 8, 15, 23 and 31 A. The analytical model overestimates L_a by 5.7% and underestimates L_u by 9.7%, so the true inductance ratio is 5.6 rather than 6.6. Its smooth-step position function also underestimates the flux linkage in partial overlap by up to 23%, because it neglects the fringing that lets flux build up before geometric overlap. Mean motoring torque is therefore underestimated by 2–17%, while peak static torque is within $\pm 10\%$. The knee model is accurate near 23 A but over-saturates at 8 A and under-saturates at 31 A. These errors exceed what the original analytical design assumed. *All drive-level results from Section V onward therefore use the FE characteristic*, imported as dense $\psi(i, \theta)$ tables with torque computed from the co-energy of the same table, so the drive model stays energy-consistent. The analytical model is retained only for the trade study of Section IX, where it is used for relative comparisons. Fig. 4 shows the flux distribution at mid-overlap, with the pole tips saturated locally.

D. Mutual Coupling

Mutual coupling was evaluated by FE with phases A and B excited simultaneously (15/15, 30/30 and 30/10 A) at seven

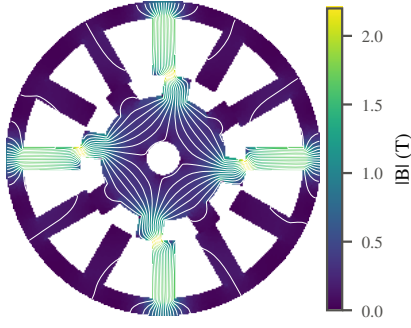


Fig. 4. FE flux density and flux lines with phase A at 30 A, 12° after the unaligned position.

TABLE VII
COUPLED THREE-PHASE FE VERSUS UNCOUPLED MODEL (24
POSITIONS PER STROKE)

Operating point	$\bar{T}$ uncoupled (N·m)	$\bar{T}$ coupled (N·m)	ΔT uncoupled (%)	ΔT coupled (%)
Launch, 80 r/min, CCC	6.64	6.69	55.7	54.2
200 r/min, 3 N·m, CCC	3.00	3.00	193.8	193.7
200 r/min, 3 N·m, cubic TSF	3.00	2.99	6.4	7.3
Rated, 1608 r/min, CCC	1.48	1.46	86.0	87.8

positions. In the commutation overlap ($\theta \leq 8.5^\circ$, when the outgoing phase is near alignment), the second phase changes ψ_A by at most 7.2%. The torque differs from the superposition of single-phase torques by at most 0.36 N·m (6.9%). The largest deviations, up to 0.65 N·m, occur for current combinations that the controller does not produce. These static tests bound the instantaneous error during phase overlap at about 5–7%.

To quantify the effect on drive-level results, the three phase currents from the (uncoupled) drive simulation were applied *simultaneously* in the FE model at 24 rotor positions over one stroke. Mutual coupling and shared-yoke saturation are then included, and the torque was compared with the uncoupled FE-table torque at the same positions (Table VII, Fig. 5). Coupling changes the mean torque by -1.5% to $+0.7\%$ and the ripple by at most 1.8 percentage points. The low ripple of the TSF is preserved (7.3% coupled against 6.4% uncoupled at these sampling points). The check is current-imposed, which is realistic under hysteresis current control at low speed. In single-pulse operation, coupling would also alter the current slightly, an effect not captured here. The uncoupled drive model is therefore adequate for the predictions in this paper.

E. Converter and Circuit Model

Each phase obeys $d\psi/dt = v - R_{ph}i$, with the converter imposing

$$v = \begin{cases} V_{dc} - 2R_{ds}i & \text{both switches on (magnetize)} \\ -(R_{ds}i + V_D) & \text{one switch on (freewheel)} \\ -(V_{dc} + 2V_D) & \text{both off, } i > 0 \text{ (demagnetize)} \end{cases} \quad (12)$$

with $R_{ds} = 4 \text{ m}\Omega$ and $V_D = 0.8 \text{ V}$. The flux equation is integrated at 40 kHz or at least 1800 steps per period, or at 50 steps per PWM period for fixed-frequency control. The current is obtained by inverting the tabulated $\psi(i, \theta)$. One phase is simulated over successive rotor pole pitches until the

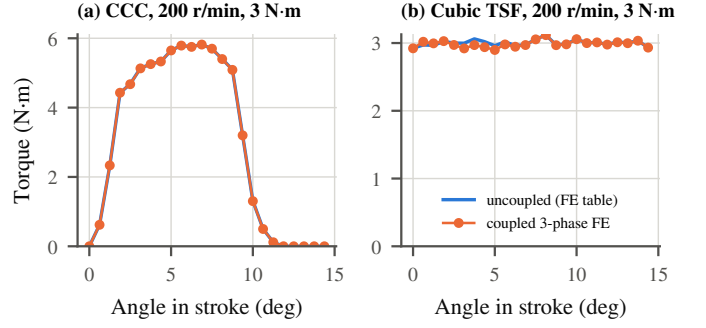


Fig. 5. Total torque over one stroke from the uncoupled FE-table model and from coupled three-phase FE solves with the same phase currents: (a) current-reference control, (b) cubic TSF, both at 200 r/min and 3 N·m.

end-of-period flux repeats to within $10^{-4}\psi_{max}$. This periodic steady state includes continuous conduction. The total torque is $T_e(\theta) = \sum_k T_{ph}(\theta - k\varepsilon)$. The DC input power equals the sum of electromagnetic power, copper loss and conduction loss to within 1.4% from 80 to 1900 r/min. The residual is the error of the explicit time integration and of the bilinear table interpolation.

F. Loss Models and Efficiency Definitions

Copper loss is $mR_{ph}I_{rms}^2$ at 100 °C. Converter conduction loss is integrated from (12). Switching energy is accumulated per transition as $\frac{1}{2}V_{dc}i t_{sw}$ ($t_{sw} = 60 \text{ ns}$) plus diode reverse recovery $Q_{rr}V_{dc}$ ($Q_{rr} = 40 \text{ nC}$). Iron loss uses the Bertotti form [25]

$$p_{fe} = k_{ns} (k_h f B^2 + k_e (fB)^2) \quad (\text{W/kg}), \quad (13)$$

calibrated to 2.7 W/kg at 1.5 T/50 Hz (M270-35A), with $k_{ns} = 1.3$ for the unipolar, non-sinusoidal SRM flux. The efficiency boundaries are:

- motor efficiency $\eta_{mot} = P_{sh}/(P_{sh} + P_{cu} + P_{fe} + P_{mech})$;
- system efficiency $\eta_{sys} = P_{sh}/P_{bat}$, from battery terminals to the motor shaft;
- converter efficiency $\eta_{conv} = (P_{bat} - P_{conv})/P_{bat}$.

The gear is excluded from all three, so battery-to-wheel efficiency is $\eta_{sys}\eta_g$.

V. CONTROL AND DESIGN FLOW

A. Commutation-Angle Optimization

The turn-on angle must lead the start of pole overlap θ_{ov} by roughly the angle needed to build up the reference current in L_u :

$$\theta_{adv} \approx \frac{\omega L_u i_{ref}}{V_{dc} - (R_{ph} + 2R_{ds})i_{ref}}. \quad (14)$$

At each speed–torque point (ω, T^*) we solve

$$\begin{aligned} & \min_{\theta_{on}, \theta_{off}, i_{ref}} P_{bat} \\ & \text{s.t. } \bar{T}_e = T^*, \quad i_{ref} \leq I_{max}, \quad \theta_{off} - \theta_{on} \leq 0.55 \tau_r, \end{aligned} \quad (15)$$

where $\tau_r = 2\pi/N_r$. The search is a grid followed by root-finding:

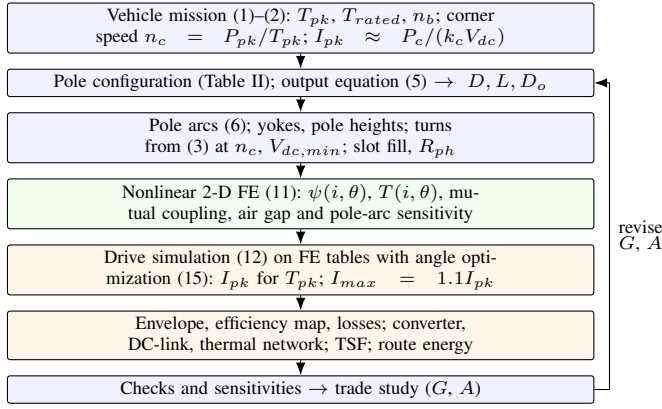


Fig. 6. Mission-to-converter design flow.

- θ_{off} takes four values at 55–95% of the interval from overlap to alignment (six values at 45–95% for maximum torque).
- θ_{on} takes three values in $[\theta_{ov} - 1.6\theta_{adv}, \theta_{ov}]$ plus four dwell-defined values ($\theta_{off} - d\tau_r$, $d = 0.35\text{--}0.5$).
- For each angle pair, i_{ref} is found by Brent's method to 0.02 A / 0.1% tolerance.

To check convergence, a $4\times$ denser grid (9 advance values, 15 θ_{off} values and 13 dwell values) improves η_{sys} by at most 1.5 points (rated point) and the maximum torque by at most 2.9% (base speed). The efficiency map is therefore slightly conservative.

The dwell limit was chosen from a sweep. At base speed, the maximum torque rises from 2.07 N·m ($dwell \leq 0.40\tau_r$) to 3.95 N·m ($0.55\tau_r$) and falls to 3.77 N·m for longer dwells. With longer dwells the tail current enters the negative-torque region. At $0.55\tau_r$ and above base speed, the optimum runs in mild continuous conduction, with a minimum current of 1.3–1.7 A (about 5% of peak), which the periodic solver handles. In a controller the solution becomes a 2-D look-up table $(\theta_{on}, \theta_{off}, i_{ref}) = f(T^*, \omega)$ [5], [26].

B. Design Flow

Fig. 6 summarizes the procedure. The smallest chopping current that gives T_{pk} with optimal angles is 26.5 A with the FE characteristic (27.9 A with the analytical one). The converter limit is set 10% higher, at $I_{max} = 29.2$ A.

VI. PREDICTED PERFORMANCE (FE-BASED)

A. Current and Torque Waveforms

Fig. 7 shows the steady-state waveforms. At launch (80 r/min), soft chopping holds the current at 29.2 A, and the mean torque is 6.63 N·m. This exceeds the 5.93-N·m requirement by 11.6%. At the rated point (1608 r/min) the turn-on angle is advanced to -2.9° and the drive operates essentially in single-pulse mode. Throughout this paper, torque ripple is the peak-to-peak value normalized to the mean,

$$\Delta T = \frac{\max_{\theta} T_e(\theta) - \min_{\theta} T_e(\theta)}{\bar{T}_e} \times 100\%, \quad (16)$$

TABLE VIII
OPTIMIZED OPERATING POINTS (FE CHARACTERISTIC)

Point	n r/min	$\bar{T}_e$ N·m	θ_{on} deg	θ_{off} deg	I_{rms} A	Ripple %	η_{mot} %	η_{sys} %
Launch	80	6.63	-0.8	21.7	20.6	60	17.7	15.9
Hill 6%	965	2.99	-0.8	15.1	10.3	101	77.5	75.3
Rated	1608	1.48	-2.9	15.1	7.7	87	80.2	77.9
Cruise	1608	0.89	-0.7	15.1	5.5	97	79.8	77.0

TABLE IX
ANALYTICAL VERSUS FE CHARACTERISTIC: KEY DRIVE RESULTS

Quantity	Analytical	FE
Current for launch torque	27.9 A	26.5 A
Launch torque at $1.1I_{pk}$	6.81 N·m	6.63 N·m
Rated η_{mot} / η_{sys}	80.9 / 78.8%	80.2 / 77.9%
Rated torque ripple	179%	87%
Route consumption	4.92 Wh/km	5.12 Wh/km

evaluated over one electrical period of the total torque T_e . With this definition a waveform that touches zero once per stroke has $\Delta T \geq 100\%$, which explains the large values for current-reference control. The ripple with current-reference control is 60% at launch and 87% at the rated point; Section VII addresses it.

The launch efficiency in Table VIII (17.7% motor, 15.9% system) is low because the output power at 80 r/min is only 54 W while the copper loss is set by the 29-A launch current. This is a short-duration, torque-dominated condition (a few seconds per start) and is not representative of steady-state riding; its thermal consequence is evaluated in Section VI-H, and the continuous operating points are the hill, rated and cruise points.

B. Torque-Speed Capability

At $I_{max} = 29.2$ A the envelope in Fig. 8 delivers 6.5–6.6 N·m up to about 600 r/min and about 3.95 N·m at 25 km/h. The available power exceeds the 500-W target above about 750 r/min. All duty points of Table I lie inside the envelope. A steady 8% climb without rider input can be sustained up to about 23 km/h, which is consistent with the pedelec concept.

C. Efficiency and Losses

Fig. 9 shows the system-efficiency map, with a maximum of 78.3% (80.6% motor efficiency). Table VIII lists the optimized operating points, and Table IX compares key results from the analytical and FE characteristics. Copper loss dominates everywhere (Fig. 10), which is the characteristic SRM penalty of carrying the magnetizing current as unipolar pulses. At the rated point, iron loss is 30% of the 64.6-W total and converter conduction loss 13%. The broader FE torque profile roughly halves the predicted ripple.

D. Switching-Frequency Sensitivity

The hysteresis controller switches at an average of 1.2–5.7 kHz, which is audible. We therefore re-simulated the hill, rated and launch points with fixed-frequency peak-current-mode PWM (soft chopping) at 4–20 kHz. For each case the

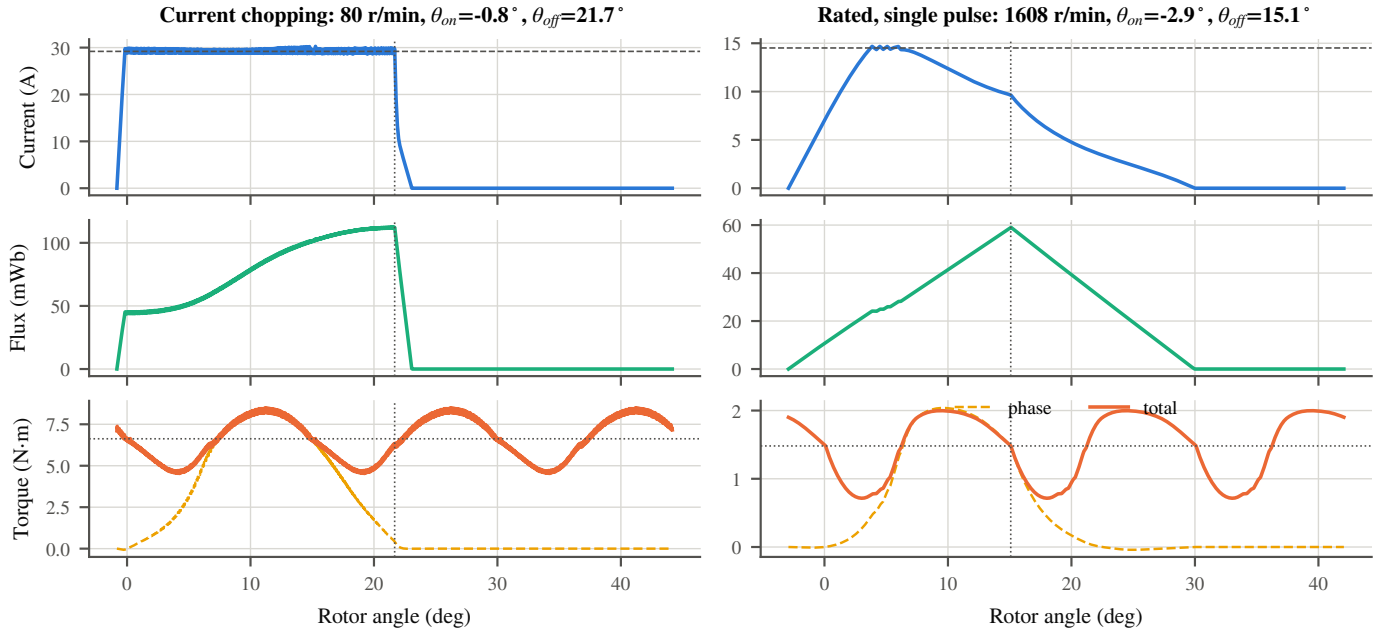


Fig. 7. Steady-state phase current, flux linkage and torque (single phase and total) at launch (left, current chopping) and at the rated point (right), computed with the FE characteristic. Dotted vertical lines mark θ_{off} .

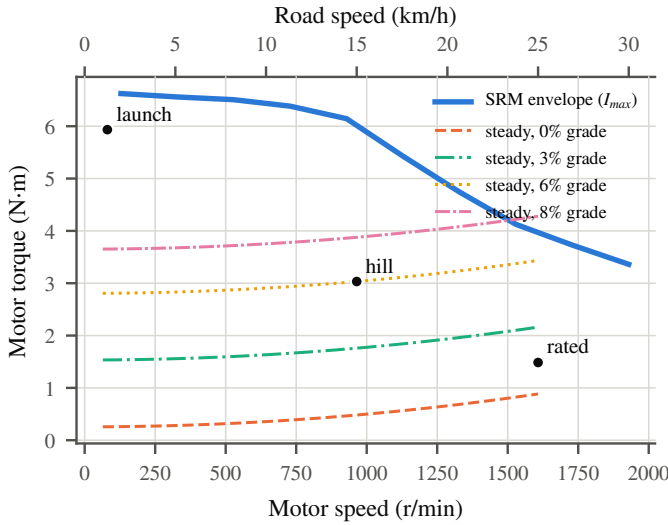


Fig. 8. FE-based torque envelope at I_{max} , steady road-load torque for 0–8% grades, and the duty points of Table I.

reference was re-solved for the same torque, and switching energy was accumulated per transition. Table X shows that at 36 V switching loss stays below 1.9 W even at 20 kHz during launch, and η_{sys} changes by less than 0.1 point. At 36 V the converter loss is conduction-dominated, so the production choice of ≥ 16 -kHz PWM does not change the efficiency conclusions. Even with four times slower switching edges, the launch switching loss would stay below 8 W.

E. Battery-Voltage Sensitivity

The 10S pack spans 30–42 V between cut-off and full charge, and the turns were chosen at 30 V. Table XI repeats the

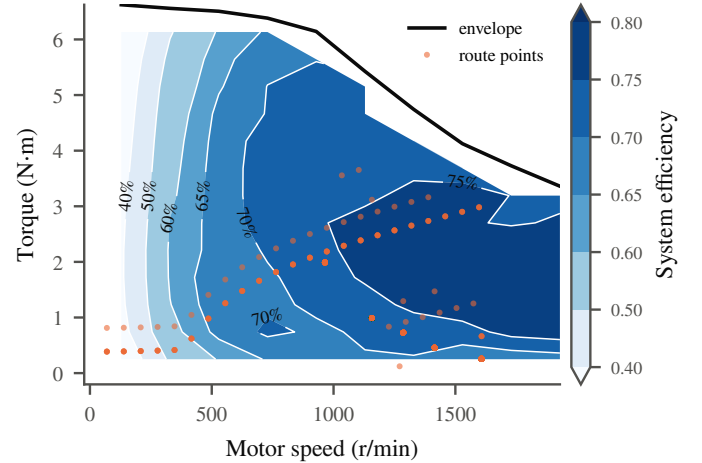


Fig. 9. FE-based system-efficiency map (η_{sys} , battery terminals to motor shaft; optimized angles at each point) with the torque envelope and the route operating points.

TABLE X
FIXED-FREQUENCY PWM: SWITCHING LOSS AND SYSTEM EFFICIENCY

	Hyst.	4 kHz	8 kHz	16 kHz	20 kHz
Launch P_{sw} (W)	0.55	0.38	0.75	1.51	1.88
Launch η_{sys} (%)	15.9	16.0	15.9	15.9	15.9
Hill P_{sw} (W)	0.07	0.02	0.02	0.06	0.09
Hill η_{sys} (%)	75.3	75.3	75.3	75.4	75.3
Rated P_{sw} (W)	0.09	0.03	0.03	0.07	0.09
Rated η_{sys} (%)	77.9	77.8	77.9	77.9	78.0

FE-based analysis at 30, 36 and 42 V with the same converter limit.

- The launch current (26.5 A) and launch torque at I_{max} (6.63 N·m) do not depend on voltage, because back-EMF

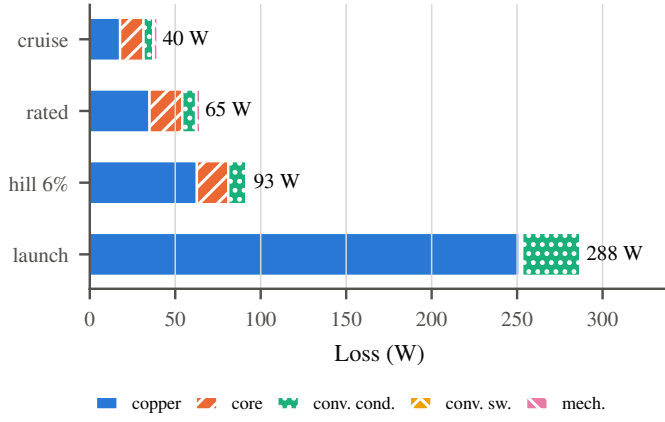


Fig. 10. Loss breakdown at the operating points of Table VIII.

TABLE XI
DC-LINK VOLTAGE SENSITIVITY (FE CHARACTERISTIC, $I_{max} = 29.2$ A)

V_{dc} (V)	30	36	42
Current for launch torque (A)	26.5	26.5	26.5
Launch torque at I_{max} (N·m)	6.63	6.63	6.64
Max. torque at 25 km/h (N·m)	2.91	3.95	4.59
Max. power at 25 km/h (W)	490	664	772
Speed at which $T_{max} = T_{rated}$ (km/h)*	42	55	68
Rated η_{mot} / η_{sys} (%)	79.3 / 77.0	80.2 / 77.9	80.9 / 78.8
Peak η_{sys} (%)	77.5	78.3	79.0

*Electrical limit only; assistance is cut at 25 km/h.

is negligible at 80 r/min.

- The voltage-limited capability does depend on voltage. At 25 km/h the maximum torque falls from 4.59 N·m at 42 V to 2.91 N·m at 30 V (490 W). This is still about twice the rated torque, so full assistance up to the cut-off is preserved at end of discharge.
- Rated efficiency changes by less than one point per 6 V.

To capture voltage sag during riding, the route model of Section VIII was extended with a battery equivalent circuit. The terminal voltage is

$$v_b = \frac{1}{2} \left(E(\text{SOC}) + \sqrt{E(\text{SOC})^2 - 4R_b P_b} \right), \quad (17)$$

where $E(\text{SOC})$ is a representative 10S NMC open-circuit-voltage curve (30 V empty, 41.8 V full; assumed rather than taken from a specific product), $R_b = 0.15 \Omega$ and P_b is the drive's battery power. The drive's efficiency and torque envelope are interpolated linearly in v_b between the three FE-based maps, and SOC is integrated from the battery current. The results in Section VIII show that, along the missions, sag changes consumption at the battery terminals by less than 3%. Internal battery loss adds 1.5–4%. On the aggressive mission at 20% SOC the minimum loaded voltage reaches 29.8 V, which touches the 30-V controller cut-off.

F. Iron-Loss Model Uncertainty

The iron-loss model carries an estimated $\pm 30\%$ uncertainty. Table XII scales it by 0.7 and 1.3 and re-optimizes the operating points.

- Rated efficiency moves by about ± 2 points (η_{mot} 78.1–82.2%, η_{sys} 75.9–79.9%).

TABLE XII
IRON-LOSS MODEL SENSITIVITY (FE CHARACTERISTIC, 36 V)

Iron-loss factor	0.7	1.0	1.3
Rated η_{mot} / η_{sys} (%)	82.2 / 79.9	80.2 / 77.9	78.1 / 75.9
Hill η_{mot} / η_{sys} (%)	79.0 / 76.7	77.5 / 75.3	76.0 / 73.8
Cruise η_{mot} / η_{sys} (%)	82.2 / 79.4	79.8 / 77.0	77.3 / 74.6
Peak η_{sys} (%)	80.4	78.3	76.4
Flat / hilly / aggr. (Wh/km)	2.35 / 4.92 / 10.66	2.47 / 5.12 / 10.92	2.60 / 5.33 / 11.19

- Route consumption moves by -4% to $+5\%$.

Iron loss is therefore the second-largest uncertainty after the analytical-model error that FE removed. A calibrated loss model, from measurement or from FE with time-domain flux waveforms, would tighten the headline efficiency to about ± 1 point.

G. Converter and DC-Link Design

The asymmetric half bridge (two switches and two diodes per phase) is the only common SRM converter that provides $+V_{dc}$, 0 and $-V_{dc}$ independently in every phase [2], [3]. These are the states that soft chopping, fast demagnetization and TSF require. With the FE characteristic, the ratings are as follows:

- device voltage $\geq 1.5V_{bat,max} = 63$ V, so 100-V MOSFETs are selected;
- $I_{max} = 29.2$ A, with a device RMS current of 20.6 A at launch;
- average battery current of 8.1 A at the rated point and 9.4 A at launch.

The DC-link capacitor is designed in two steps. The first-cut charge bound assumes that the battery supplies only the mean current:

$$C \geq \frac{\Delta Q}{\Delta V}, \quad q(t) = \int_0^t (i_{dc} - \bar{i}_{dc}) d\tau, \quad (18)$$

$$\Delta Q = \max_t q(t) - \min_t q(t).$$

At the rated point (stroke frequency 643 Hz), (18) gives 1.56 mF for 5% ripple. At launch the stroke frequency is only 32 Hz and the bound becomes a meaningless 30 mF, because in reality the battery supplies most of that low-frequency ripple. The second step therefore resolves $i_{dc}(t)$ into harmonics and divides each one between the capacitor branch $Z_c = R_{ESR} + 1/(j\omega C)$ and the battery branch $Z_b = R_b + j\omega L_b$. We assume $R_{ESR} = 20$ m Ω , $R_b = 0.15 \Omega$ (10S pack) and $L_b = 1 \mu\text{H}$ (cable). Table XIII gives the results:

- At the rated point, 1 mF keeps the link ripple at 1.7 V peak to peak (4.8%) with 4.2 A RMS in the capacitor.
- At launch, the 4–9-V peak-to-peak ripple at 32 Hz is set mainly by the battery impedance and is barely affected by C .
- The capacitor rating is governed by its ripple current (13.4 A RMS at launch), not by capacitance.

A 1–2.2-mF low-ESR bank rated for at least 14 A RMS is therefore sufficient. The battery also carries 4–7 A RMS of low-frequency ripple at launch. With $R_b = 0.15 \Omega$ this adds 2.9–6.9 W of internal heating for the few seconds of a launch. Whether this ripple is acceptable over the pack life

TABLE XIII
DC-LINK CURRENT SHARING ($R_{ESR} = 20 \text{ m}\Omega$, $R_b = 0.15 \Omega$, $L_b = 1 \mu\text{H}$)

	$C' (\mu\text{F})$	470	1000	2200	3800
Rated	$I_{C,rms} \text{ (A)}$	3.9	4.2	4.5	4.6
	$I_{bat,ac} \text{ (A)}$	3.3	2.6	1.7	1.2
	$\Delta V_{pp} \text{ (V)}$	2.2	1.7	1.2	0.9
Launch	$I_{C,rms} \text{ (A)}$	13.4	13.4	13.4	13.4
	$I_{bat,ac} \text{ (A)}$	6.8	5.5	4.8	4.4
	$\Delta V_{pp} \text{ (V)}$	9.0	7.2	5.6	4.4

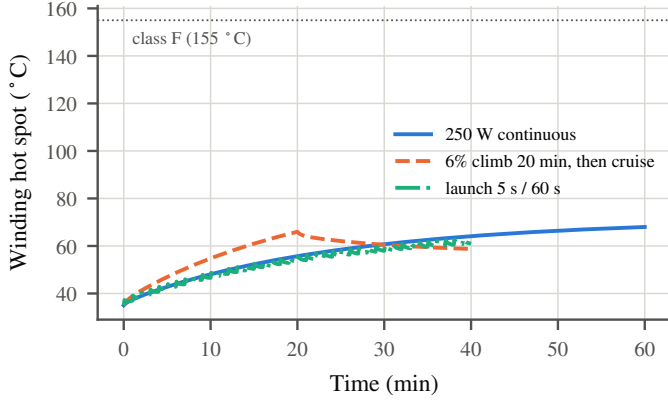


Fig. 11. Predicted winding hot-spot temperature ($h = 25 \text{ W/m}^2\text{K}$, 35°C ambient).

must be checked against the cell manufacturer's ripple-current specification, which is outside the scope of this paper.

H. Predicted Thermal Envelope

The temperatures in this subsection are predictions of an uncalibrated lumped-parameter model and define a predicted thermal envelope, not validated temperature rises. A four-node network replaces the single-node estimate of the first version. The nodes are the winding, stator, rotor and hub shell. The conductances are estimated from the geometry:

- winding to stator: 25 W/K (slot liner, 0.35 mm , $0.25 \text{ W/m}^2\text{K}$);
- stator to shell: 23 W/K (contact, $1500 \text{ W/m}^2\text{K}$);
- rotor to shell: 0.6 W/K ;
- shell to ambient: hA_{sh} , with $A_{sh} = 0.075 \text{ m}^2$.

The heat capacities are 463, 708, 353 and 990 J/K . Copper loss is temperature dependent, and the hot spot is taken 15% above the mean winding rise. Fig. 11 shows three scenarios at $h = 25 \text{ W/m}^2\text{K}$ and 35°C . Table XIV gives the sensitivity to h and ambient temperature, including steady-state values, because the shell time constant is about 9 min and the full network settles over more than 1 h. The rated point is comfortable in all cases (at most 107°C). A sustained 6% climb at low airflow ($h = 15 \text{ W/m}^2\text{K}$, 15 km/h) and 45°C ambient reaches 142°C at steady state, only 13 K below the class-F limit. For hot-climate or mountainous use we therefore recommend class-H insulation or a thermal derating of the hill-assist torque.

TABLE XIV
PREDICTED WINDING HOT SPOT ($^\circ\text{C}$): SENSITIVITY TO h AND AMBIENT

$h \text{ (W/m}^2\text{K)}$	15		25		40	
$T_{amb} \text{ (}^\circ\text{C)}$	35	45	35	45	35	45
Rated, 60 min	80.9	91.9	68.0	78.7	58.0	68.5
Rated, steady state	95.5	106.8	71.4	82.2	58.6	69.1
6% climb, 20 min	70.4	81.4	66.0	76.9	61.0	71.7
6% climb, steady state	129.4	142.0	90.0	101.5	70.3	81.2
Launch 5 s / 60 s, 40 min	70.3	81.2	62.9	73.7	56.2	66.8

TABLE XV
CURRENT-REFERENCE CONTROL (CCC) VERSUS TSF (FE CHARACTERISTIC)

$n \text{ (r/min)}$	$T^* \text{ (N-m)}$	Control	Ripple (%)	$I_{rms} \text{ (A)}$	$\eta_{sys} \text{ (%)}$
200	3.0	CCC	196.5	9.33	48.3
		Linear TSF	16.1	12.97	32.2
		Cubic TSF	10.4	11.86	35.9
		Exp. TSF	17.1	10.84	39.9
200	1.5	CCC	198.7	6.08	49.2
		Cubic TSF	13.4	7.06	39.9
500	3.0	CCC	146.1	9.64	66.7
		Cubic TSF	32.8	12.31	54.6
800	1.5	CCC	83.4	6.74	70.1
		Cubic TSF	39.2	7.53	63.3

800 r/min/3.0 N-m, 1200/1.5, 1608/1.48: TSF not feasible within I_{max}

VII. TORQUE-RIPPLE REDUCTION WITH TORQUE-SHARING FUNCTIONS

A TSF assigns each phase a torque reference $T_k^* = T^* f_k(\theta)$. The rising function of the incoming phase and the falling function of the outgoing phase overlap over θ_{ov} and always sum to one [11], [27]. We implemented linear, cubic and exponential shapes. Each phase reference is converted to a current reference through the inverse FE static characteristic $i(T, \theta)$ and tracked by a three-level hysteresis controller ($+V_{dc}$, 0 , $-V_{dc}$). For each shape, the turn-on angle ($3\text{--}7.5^\circ$) and overlap ($1.5\text{--}4.5^\circ$) were chosen for minimum ripple, and the torque command was trimmed so that the mean torque equals the target.

Table XV and Fig. 12 show the results. Below 500 r/min, the cubic TSF reduces the peak-to-peak ripple from about 197% to 10–13% at 200 r/min and to 33% at 500 r/min. At 800 r/min and 1.5 N-m the ripple falls from 83% to 39%. The price is a 12–28% higher RMS current and 7–12 points of system efficiency. Above about 800 r/min, or 3 N-m at 800 r/min, no TSF solution exists within I_{max} at 36 V, because the back-EMF prevents the falling phase from following its reference. This is the well-known speed limit of TSF control [27], [28], and it is particularly restrictive at low DC-link voltage. A practical controller is therefore hybrid: TSF for launch, walk-assist and low-speed climbing, where ripple is felt most and power is small, and current-reference angle control above about 12 km/h. Coupled three-phase FE solves (Section IV-D) confirm that the TSF ripple reduction survives mutual coupling ($6.4\% \rightarrow 7.3\%$ at the 200-r/min point).

VIII. ROUTE ENERGY EVALUATION

Three standardized missions are simulated quasi-statically:

- *flat urban*: 6.4 km on level roads with six stops;

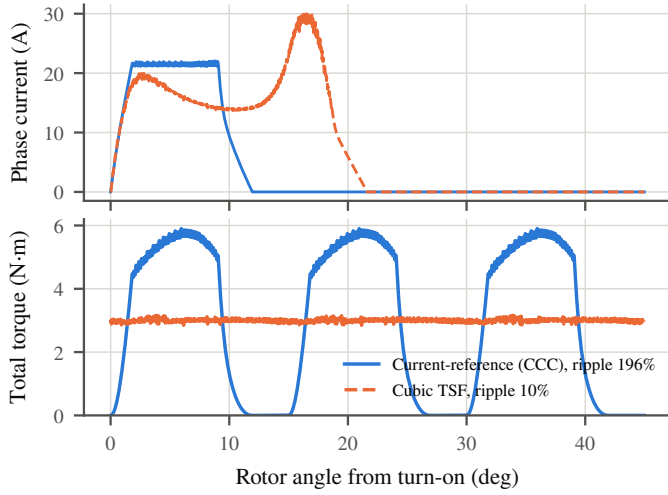


Fig. 12. Phase current and total torque at 200 r/min and 3 N-m: current-reference control versus cubic TSF.

TABLE XVI

STANDARDIZED MISSIONS: PREDICTED CONSUMPTION AND ESTIMATED RANGE

Mission	Flat urban	Hilly urban	Aggressive
Distance (km)	6.4	6.4	5.4
Fixed 36 V: Wh/km	2.47	5.12	10.92
Fixed 36 V: est. range, 90% of 360 Wh (km)	131	63	30
Battery model, SOC 100%: Wh/km (terminal)	2.40	5.08	10.89
Battery model, SOC 20%: Wh/km (terminal)	2.47	5.10	10.95
Battery model, SOC 50%: Wh/km (incl. R_b loss)	2.50	5.23	11.28
Min. loaded voltage at SOC 20% (V)	32.0	31.4	29.8
Est. range to 30-V cut-off with sag (km)	142	68	32

- *hilly urban*: 6.4 km with five stops, a 700-m climb at 6% and a 900-m descent at -4% (Fig. 13);
- *aggressive*: 5.4 km with climbs of 6–10% totalling 2.2 km and descents at -4% and -7% .

In all three, the rider supplies up to 100 W, force-limited at low speed. The motor supplies the remaining tractive power below 25 km/h, with efficiency interpolated from the FE-based maps, and braking is mechanical.

On the hilly mission (18.9 min), the predictions at a fixed 36 V are as follows:

- the rider contributes 25.5 Wh and the motor 23.4 Wh at the shaft;
- the motor draws 32.7 Wh from the battery, a cycle-average η_{sys} of 71.4%;
- consumption is 5.12 Wh/km.

Table XVI compares the three missions, with and without the battery equivalent circuit (17). Consumption varies more than fourfold between missions, from 2.5 to 10.9 Wh/km, so a single route figure must not be read as a general consumption prediction. With sag, the *estimated* range to the 30-V cut-off is 142, 68 and 32 km. These figures use nearly the full capacity, whereas the fixed-voltage estimates assume 90%. The torque demand never exceeds the envelope, even at 20% SOC.

The brakes dissipate 7.2 Wh at the wheel. With a generating efficiency of about 0.75 and $\eta_g = 0.95$, regeneration could return at most about 5 Wh, roughly 15% of the battery energy on the hilly mission and less on the flat one. The

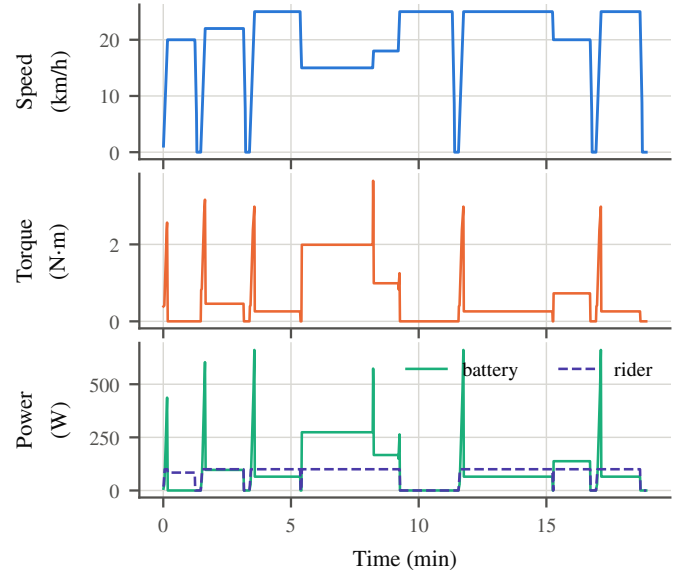


Fig. 13. Hilly urban mission: speed, motor torque, and battery and rider power.

TABLE XVII

GEAR-MODEL SENSITIVITY (HILLY MISSION, FIXED 36 V)

Gear model	Launch T_m (N-m)	Wh/km	Est. range (km)
$\eta_g = 0.90$	6.26	5.38	60
$\eta_g = 0.93$	6.06	5.22	62
$\eta_g = 0.95$ (baseline)	5.93	5.12	63
$\eta_g = 0.97$	5.81	5.02	65
$\eta_g = 0.97 + 3$ W drag	5.83	5.15	63
$\eta_g = 0.95 + 5$ W drag	5.96	5.33	61

asymmetric bridge supports regeneration without hardware change. It is omitted in the baseline because it complicates the brake interface and the freewheel arrangement for a modest gain.

The quasi-static model neglects electrical and mechanical transients, controller delays and the thermal state. Battery sag is included only in the results labeled “battery model”. Table XVII reports the sensitivity to the gear model. Over the plausible range $\eta_g = 0.90$ –0.97, consumption varies from 5.02 to 5.38 Wh/km (estimated range 65–60 km), and the launch torque requirement from 5.81 to 6.26 N-m. The launch requirement stays within the envelope in all cases, but at $\eta_g = 0.90$ the margin shrinks to 6%.

IX. DESIGN TRADE-OFFS AND TEST OF THE CORNER-SPEED RULE

Table XVIII and Fig. 14 vary the gear ratio G and the peak electric loading A . Each machine is redesigned and the current is re-sized. The analytical characteristic is used here for computational reasons; it reproduces the FE rated efficiency of the baseline to within 1 point (Table IX), so the trends are reliable even though absolute values carry the uncertainty of Table VI. Three observations stand out.

- 1) The required peak current (for the launch torque, before the 10% converter margin) lies between 25.2 and 30.6 A across a 3.9-fold range of active mass. Fig. 15(a) shows

TABLE XVIII
TRADE STUDY: GEAR RATIO G AND PEAK ELECTRIC LOADING A
(ANALYTICAL CHARACTERISTIC)

G	A kA/m	D_o mm	L mm	Mass kg	I_{max} A	η_{mot} %	η_{sys} %
5	60	143.8	38.1	4.10	27.7	75.0	73.2
5	45	158.3	41.9	5.51	29.2	81.7	79.8
5	35	172.1	45.6	7.14	31.2	84.2	81.9
8	60	122.9	32.6	2.53	29.3	76.9	75.1
8	45	135.3	35.9	3.40	30.7	80.9	78.8
8	35	147.1	39.0	4.40	32.0	82.7	80.3
11	60	110.6	29.3	1.82	30.3	77.4	75.7
11	45	121.7	32.2	2.45	31.6	79.8	77.6
11	35	132.3	35.1	3.17	33.7	80.9	78.2

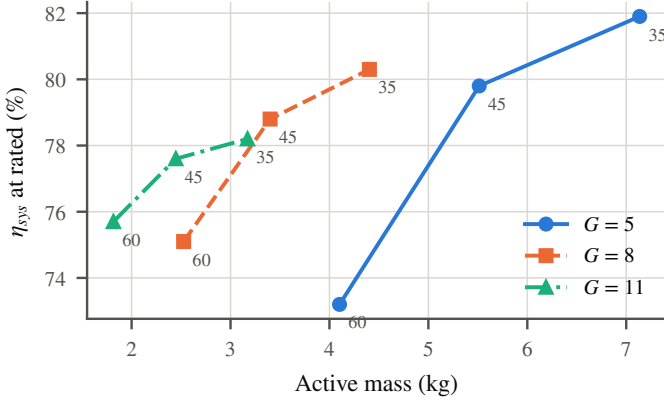


Fig. 14. Rated-point system efficiency versus active mass for gear ratios 5, 8 and 11. The labels give the peak electric loading in kA/m.

this. A linear fit of current against mass has a slope of -0.06 A/kg ($-0.2\%/kg$) and $R^2 = 0.004$, so the current is statistically independent of machine size. The loop-utilization factor is $k_c = 0.60 \pm 0.03$ (mean $\pm$ standard deviation, range 0.54–0.66), and the rule $I_{pk} = P_c/(\bar{k}_c V_{dc,min}) = 27.8$ A reproduces every design to within 10.3%. To test the voltage dependence, the baseline frame ($G = 8$, $A = 45$ kA/m) was redesigned for 24-, 36- and 48-V packs ($V_{dc,min} = 20, 30$ and 40 V; 27, 41 and 54 turns per pole). The required current falls from 42.3 to 27.9 to 21.2 A, a log–log slope of -1.002 (Fig. 15(b)), with $k_c = 0.59$ – 0.60 in all three cases. Both tests support (4) within the scope of this design family (one-stroke dwell, same pole configuration and loading coefficients).

- 2) Rated efficiency is governed mainly by electric loading, i.e., by active mass. For $G = 8$, lowering A from 60 to 35 kA/m raises η_{sys} from 75.1% to 80.3% while active mass grows from 2.5 to 4.4 kg.
- 3) At similar mass, the gear ratio matters little. The $G = 8/45$ kA/m baseline (3.40 kg, 78.8%) and the $G = 11/35$ kA/m variant (3.17 kg, 78.2%) are within one point. The ratio should therefore be chosen on gear cost, noise and diameter. The single-stage 8:1 baseline favors cost, and the 11:1 variant gives the smallest diameter for a given efficiency.

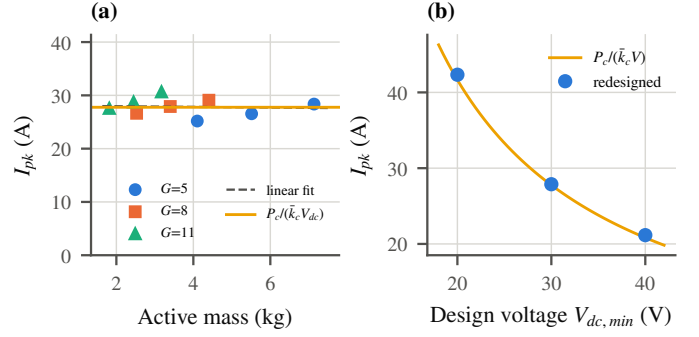


Fig. 15. Test of the corner-speed rule (4). (a) Required peak current versus active mass for the nine designs of Table XVIII, with a linear fit and the rule prediction $P_c/(\bar{k}_c V_{dc,min})$. (b) Required current of the baseline frame redesigned for 24-, 36- and 48-V packs, against $P_c/(\bar{k}_c V)$.

X. DISCUSSION

A. Position Sensing

Commutation needs rotor position to about 1° mechanical. Three Hall sensors on a target ring give 15° edges. Between edges, the angle can be extrapolated using speed and acceleration estimated from the edge timing. This is adequate when the speed change between edges is small, which holds above about 50 r/min given the vehicle inertia. The quantization error and the sensor-alignment tolerance (typically $\pm 1^\circ$) appear directly as commutation-angle errors, and their effect follows from the sensitivity of the optimized operating points to θ_{off} . At standstill and during a hill start the Hall states only identify a 15° sector, so the controller must excite the phase pair that is known to give positive torque throughout that sector and switch to interpolation once two edges have been seen. Direction reversal (walking backward) is detected from the Hall state sequence. Sensorless schemes [29] are attractive for sealed hubs but must achieve reliable stand-still starts on a grade, which is exactly the launch condition that sizes the drive.

B. Air Gap and Mechanical Tolerances

The FE results in Table XIX quantify the air-gap sensitivity. At 0.35 mm the mean torque at 30 A drops by 4.0% (8.3% at 15 A) and L_a by 10%. At 0.40 mm the torque drops by 8.2% (16% at 15 A). Table XX gives an *assumed* tolerance allocation that is typical for small motors and must be confirmed for a specific design. Worst-case stacking reduces the local gap to 0.215 mm, and the root-sum-square (RSS) combination to 0.26 mm. This corresponds to a 13–28% eccentricity with corresponding unbalanced magnetic pull, so the rotor bearings and the axle must be dimensioned for it. Because the wheel loads in a geared hub are carried by the hub bearings rather than by the motor bearings, the remaining contributors are manufacturing tolerances.

C. Equal-Envelope Benchmark Against a Surface-PM Machine

Catalog efficiencies of PM e-bike motors are not comparable with the present predictions, because their measurement

TABLE XIX
FE AIR-GAP SENSITIVITY (SAME TURNS)

g (mm)	T_{15A}	T_{30A}	L_a (mH)	L_u (mH)	L_a/L_u
0.30	1.80	4.52	8.56	1.52	5.64
0.35	1.65	4.34	7.69	1.50	5.11
0.40	1.51	4.15	6.99	1.50	4.67

TABLE XX
ASSUMED AIR-GAP TOLERANCE ALLOCATION (DESIGN BUDGET, NOT MEASURED)

Contributor	Allocation (mm)
Bearing radial clearance (small deep-groove, CN)	0.015
Bearing-seat / housing concentricity	0.020
Rotor stack outer-diameter runout	0.020
Stator bore tolerance (punching, stacking)	0.020
Axle/rotor deflection under magnetic pull	0.010
Worst-case sum / RSS	0.085 / 0.039
Minimum local gap, worst case / RSS	0.215 / 0.261

TABLE XXI
EQUAL-ENVELOPE BENCHMARK: ANALYTIC SPM VERSUS SRM

	SPM (analytic)	SRM (analytic)	SRM (FE)
Active mass / magnet mass (kg)	3.04 / 0.16	3.40 / 0	3.40 / 0
Launch current (A, peak)	37.5	27.9	26.5
Launch η_{mot} / η_{sys} (%)	66.0 / 57.6	19.9 / 18.0	17.7 / 15.9
Hill η_{mot} / η_{sys} (%)	94.5 / 93.5	75.9 / 73.6	77.5 / 75.3
Rated η_{mot} / η_{sys} (%)	90.6 / 90.2	80.9 / 78.8	80.2 / 77.9
Cruise η_{mot} / η_{sys} (%)	84.9 / 84.5	79.8 / 77.0	79.8 / 77.0
Flat / hilly / aggr. (Wh/km)	3.03 / 4.93 / 8.68	—	2.47 / 5.12 / 10.92
Coasting loss, no clutch, 25 km/h (W)	19.7	0	0

boundaries, voltages, temperatures and test methods differ or are not stated. We therefore sized a first-order analytic surface-PM (SPM) benchmark to the *same* constraints:

- the same mission, envelope ($D_o = 135.3$ mm, $L = 35.9$ mm), gear and battery;
- the same MOSFETs, iron-loss coefficients and mechanical loss;
- the same efficiency boundary and route model.

The benchmark is a 12-slot/10-pole concentrated-winding machine (winding factor 0.933) with N42SH magnets ($B_r = 1.17$ T at 100 °C, 3 mm thick, 0.83 pole arc) and a 0.5-mm air gap. Its turns follow the same voltage logic as the SRM: the no-load line-to-line EMF at 25 km/h equals the 30-V minimum battery voltage. It is driven by sinusoidal $I_d = 0$ control from a six-switch inverter at 16 kHz. Its iron loss uses the same Bertotti coefficients ($k_{ns} = 1$), and an assumed magnet eddy-current loss of 10% of the stator iron loss is added. The model neglects saturation, slotting harmonics and field weakening. It therefore matches the fidelity of the SRM's *analytical* model, and Table XXI lists both SRM models.

The benchmark gives an unambiguous result, which we report as it stands. At the same envelope, the SPM is predicted to be about 11–12 points more efficient at the rated point and far better at launch, where the SRM's copper loss is ten times higher. It also uses 4% and 20% less energy on the hilly and aggressive missions. On the flat, light-assist mission the SRM uses 18% less energy. The motor then delivers small torques for long periods, and the SPM's magnet-driven iron loss (about 18 W at 25 km/h, independent of load) dominates,

while the SRM's iron loss scales with excitation. The SPM also dissipates about 20 W when coasting without a freewheel clutch. Geared hubs normally include such a clutch, but direct-drive hubs do not. Against these efficiency deficits, the SRM offers zero magnet content (0.16 kg of NdFeB avoided), robustness to temperature and overspeed, phase-independent fault tolerance and a simpler rotor. Its case for pedelecs therefore rests on cost, supply-chain and robustness grounds and on light-assist use, not on peak efficiency. Because the SPM model is first-order, its efficiencies are likely optimistic by a few points, but not by enough to reverse the comparison at the rated point.

D. Limitations and Validation Status

- *No experimental results.* All performance, efficiency, range and temperature values are model predictions. Prototype dynamometer tests (phase voltage and current, torque and ripple, DC input power, efficiency map, temperature rise and acoustic noise) are the next step of this work.
- *Magnetics.* 2-D FE neglects end effects except for the analytical end-winding term. Axial fringing in a 36-mm stack typically adds a few percent of inductance. Mutual coupling is not included in the drive model; coupled FE solves bound its effect on mean torque at 1.5% (Section IV-D).
- *Losses.* Iron loss uses a peak-flux-density model with a non-sinusoidal factor ($\pm 30\%$).
- *Thermal network.* The thermal parameters are geometry-based estimates that have not been calibrated.
- *Route model.* The route model is quasi-static; battery sag is included through an equivalent circuit (Section VI-E) but electrical and mechanical transients are not.
- *Acoustics.* Acoustic noise and radial-force vibration are outside the present modeling scope, and no claim is made that the design is acoustically acceptable.

XI. CONCLUSION

This paper has presented an analytical, FE and simulation-based design methodology for a switched reluctance motor drive for a 250-W, 36-V pedelec. All results are model predictions; no prototype has been measured.

- *Corner-speed rule.* The phase turns are chosen at the mission-derived corner speed $n_c = P_{pk}/T_{pk}$. For the specified 36-V, 500-W corner power, this approximately halves the peak current relative to sizing the turns for launch torque at 25 km/h. The relation $P_c = k_c V_{dc} I_{pk}$ recasts the classical converter volt-ampere argument for sizing. Within this design family it holds with $k_c = 0.60 \pm 0.03$, shows no dependence of current on machine mass ($R^2 = 0.004$ over a 3.9-fold mass range), and gives $I_{pk} \propto V^{-1.00}$ across 24–48-V designs.
- *FE-based numerical verification.* Nonlinear 2-D FE showed that the analytical model errs by up to 23% in flux linkage and 17% in mean torque, so FE tables were used for all drive-level predictions. Coupled three-phase FE solves changed mean torque by at most 1.5%.

- *Predicted performance.* The 12/8, $\varnothing$ 135-mm, 3.4-kg (active) machine with an 8:1 gear and a 100-V MOSFET asymmetric bridge meets the launch torque at 26.5 A with an 11.6% margin at any battery voltage from 30 to 42 V. It reaches 80.2% motor and 77.9% system efficiency at the rated point (± 2 points for $\pm 30\%$ iron-loss uncertainty), and consumes 2.5–10.9 Wh/km on flat to aggressive missions.
- *Torque ripple.* TSF control reduces ripple to 10–33% below 500 r/min but is infeasible above about 800 r/min at 36 V, which motivates a hybrid controller.
- *Converter and thermal.* Switching losses are negligible at 36 V up to 20 kHz, and the DC link is ripple-current-limited. In the predicted thermal envelope, only sustained climbing in hot climates approaches the insulation limit.
- *PM benchmark.* An equal-envelope analytic SPM is predicted to be about 12 points more efficient at the rated point and to use less energy on hilly and aggressive missions. The SRM uses 18% less energy on flat light-assist riding and needs no magnets.

The SRM is therefore a credible magnet-free option for pedelecs whose case rests on cost, supply chain, robustness and light-assist use rather than on peak efficiency. Prototype validation of these predictions is the subject of ongoing work.

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